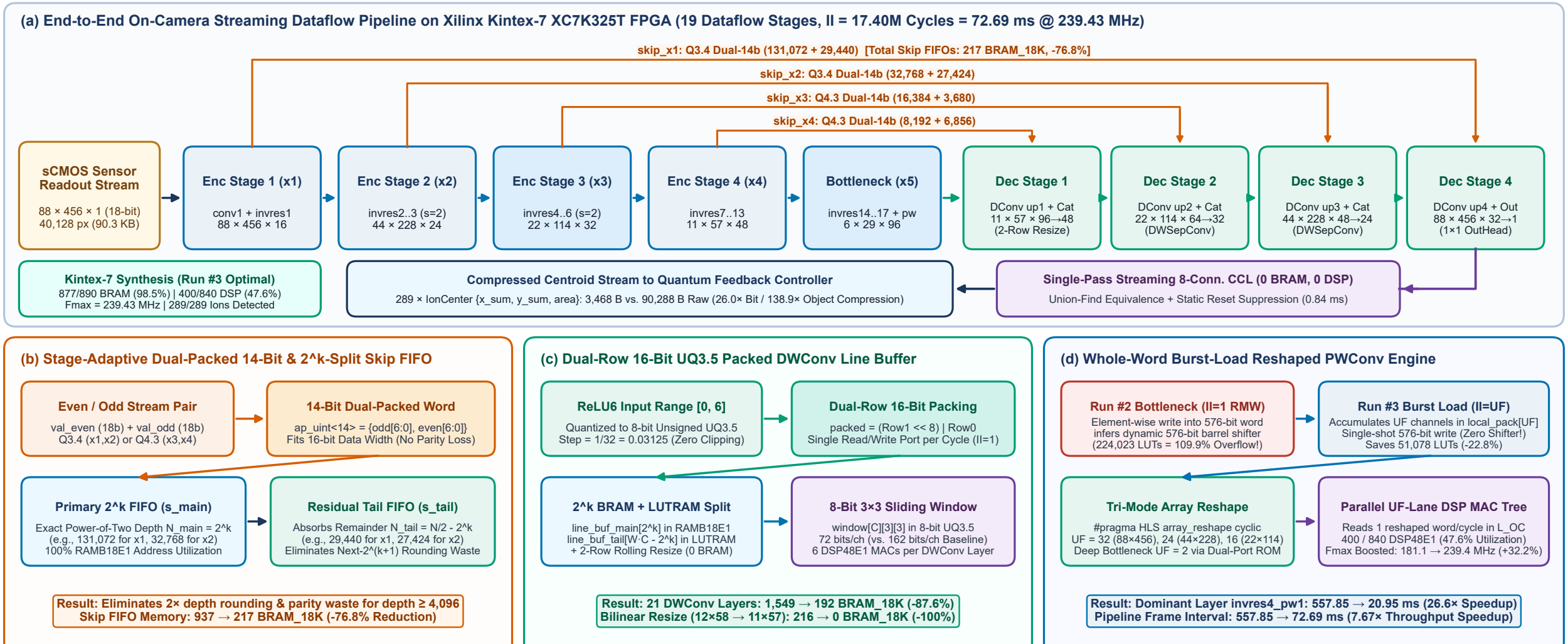

(a) End-to-End On-Camera Streaming Dataflow Pipeline on Xilinx Kintex-7 XC7K325T FPGA (19 Dataflow Stages, II = 17.40M Cycles = 72.69 ms @ 239.43 MHz)
skip_x1: Q3.4 Dual-14b (131,072 + 29,440) [Total Skip FIFOs: 217 BRAM_18K, -76.8%]
skip_x2: Q3.4 Dual-14b (32,768 + 27,424)
skip_x3: Q4.3 Dual-14b (16,384 + 3,680)
skip_x4: Q4.3 Dual-14b (8,192 + 6,856)
sCMOS Sensor Readout Stream
88 × 456 × 1 (18-bit)
40,128 px (90.3 KB)
Enc Stage 1 (x1)
conv1 + invres1
88 × 456 × 16
Enc Stage 2 (x2)
invres2..3 (s=2)
44 × 228 × 24
Enc Stage 3 (x3)
invres4..6 (s=2)
22 × 114 × 32
Enc Stage 4 (x4)
invres7..13
11 × 57 × 48
Bottleneck (x5)
invres14..17 + pw
6 × 29 × 96
Dec Stage 1
DConv up1 + Cat
11 × 57 × 96→48
(2-Row Resize)
Dec Stage 2
DConv up2 + Cat
22 × 114 × 64→32
(DWSepConv)
Dec Stage 3
DConv up3 + Cat
44 × 228 × 48→24
(DWSepConv)
Dec Stage 4
DConv up4 + Out
88 × 456 × 32→1
(1×1 OutHead)
Kintex-7 Synthesis (Run #3 Optimal)
877/890 BRAM (98.5%) | 400/840 DSP (47.6%)
Fmax = 239.43 MHz | 289/289 Ions Detected
Compressed Centroid Stream to Quantum Feedback Controller
289 × IonCenter {x_sum, y_sum, area}: 3,468 B vs. 90,288 B Raw (26.0× Bit / 138.9× Object Compression)
Single-Pass Streaming 8-Conn. CCL (0 BRAM, 0 DSP)
Union-Find Equivalence + Static Reset Suppression (0.84 ms)
(b) Stage-Adaptive Dual-Packed 14-Bit & 2^k-Split Skip FIFO
Even / Odd Stream Pair
val_even (18b) + val_odd (18b)
Q3.4 (x1,x2) or Q4.3 (x3,x4)
14-Bit Dual-Packed Word
ap_uint<14> = {odd[6:0], even[6:0]}
Fits 16-bit Data Width (No Parity Loss)
Primary 2^k FIFO (s_main)
Exact Power-of-Two Depth N_main = 2^k
(e.g., 131,072 for x1, 32,768 for x2)
100% RAMB18E1 Address Utilization
Residual Tail FIFO (s_tail)
Absorbs Remainder N_tail = N/2 - 2^k
(e.g., 29,440 for x1, 27,424 for x2)
Eliminates Next-2^(k+1) Rounding Waste
Result: Eliminates 2× depth rounding & parity waste for depth ≥ 4,096
Skip FIFO Memory: 937 → 217 BRAM_18K (-76.8% Reduction)
(c) Dual-Row 16-Bit UQ3.5 Packed DWConv Line Buffer
ReLU6 Input Range [0, 6]
Quantized to 8-bit Unsigned UQ3.5
Step = 1/32 = 0.03125 (Zero Clipping)
Dual-Row 16-Bit Packing
packed = (Row1 << 8) | Row0
Single Read/Write Port per Cycle (II=1)
2^k BRAM + LUTRAM Split
line_buf_main[2^k] in RAMB18E1
line_buf_tail[W·C - 2^k] in LUTRAM
+ 2-Row Rolling Resize (0 BRAM)
8-Bit 3×3 Sliding Window
window[C][3][3] in 8-bit UQ3.5
72 bits/ch (vs. 162 bits/ch Baseline)
6 DSP48E1 MACs per DWConv Layer
Result: 21 DWConv Layers: 1,549 → 192 BRAM_18K (-87.6%)
Bilinear Resize (12×58 → 11×57): 216 → 0 BRAM_18K (-100%)
(d) Whole-Word Burst-Load Reshaped PWConv Engine
Run #2 Bottleneck (II=1 RMW)
Element-wise write into 576-bit word
infers dynamic 576-bit barrel shifter
(224,023 LUTs = 109.9% Overflow!)
Run #3 Burst Load (II=UF)
Accumulates UF channels in local_pack[UF]
Single-shot 576-bit write (Zero Shifter!)
Saves 51,078 LUTs (-22.8%)
Tri-Mode Array Reshape
#pragma HLS array_reshape cyclic
UF = 32 (88×456), 24 (44×228), 16 (22×114)
Deep Bottleneck UF = 2 via Dual-Port ROM
Parallel UF-Lane DSP MAC Tree
Reads 1 reshaped word/cycle in L_OC
400 / 840 DSP48E1 (47.6% Utilization)
Fmax Boosted: 181.1 → 239.4 MHz (+32.2%)
Result: Dominant Layer invres4_pw1: 557.85 → 20.95 ms (26.6× Speedup)
Pipeline Frame Interval: 557.85 → 72.69 ms (7.67× Throughput Speedup)